

# Scott Nguyen

scott.nguyen8901@gmail.com |(515) 943-6283 |Security Clearance: Interim Secret

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## EDUCATION

Master of Science in Electrical Engineering | Space Systems Engineering

University of New Mexico

Spring 2026

GPA: 3.94/4.00

Master of Science in Aerospace Engineering

University of Illinois Urbana-Champaign

Fall 2024

Bachelor of Science in Aerospace Engineering

Iowa State University

Spring 2022

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## SKILLS

Programming Languages: MATLAB, Python, C/C++, Ruby

Frameworks / Libraries / Tools: NumPy, SciPy, Matplotlib, Astropy, Poliastro, bpy, git

Applications: Simulink, Blender, GRASP, SystemVue, Advanced Design System

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## WORK EXPERIENCE

Student Co-Op, Electronics for Contested Space Group

September 2025 – Present

MIT Lincoln Laboratory

- Led design of a new **S-band communication architecture** for a U.S. radar network, integrating flight-like hardware and simulation to reestablish links with lost-comm satellites.
- Implemented an **Unscented Kalman Filter (UKF)** for precise radio frequency measurements and state estimation
- Built a probabilistic detection tool to compute observation likelihoods from resident space object (RSO) properties and optical sensor performance.

Guidance, Navigation & Controls Engineer Intern

May 2025 – August 2025

Blue Canyon Technologies

- Verified functionality and polarity of IMU, Nano Star Tracker, Reaction Wheels, Torque Rods, and Sun Sensors via hardware testing and data checks
- Performed regression analysis of two-axis **Solar Array Drive Assembly (SADA)** momentum management and validated command interfaces for precise control and reliability
- Automated **SADA** validation by developing **Ruby** test scripts and mapping telemetry channels to **COSMOS**

Guidance, Navigation & Controls Engineer Intern

January 2025 – April 2025

Blue Origin

- Integrated **Active Disturbance Rejection Control (ADRC)** and **Sliding Mode Control (SMC)** to develop a robust algorithm for stabilizing nonlinear MIMO dynamics of the BE-7 engine
- Evaluated control performance by injecting disturbances and demonstrated improved accuracy in setpoint tracking
- Integrated flight software into **Simulink** using **S-functions** in **C** to enable testing and verification

Guidance, Navigation & Controls Engineer Intern

May 2024 – August 2024

Varda Space Industries

- Conducted trade studies to optimize gravity models for mission requirements and select optimal filter type (**EKF** vs. **UKF**)
- Built Monte Carlo simulations to quantify reentry uncertainty, generating latitude/longitude covariance ellipsoids and a reentry dispersion cloud for flight safety analysis and capsule recovery planning
- Implemented an **EKF** for state estimation, optimizing ground station timing for minimal residuals and precise delta-v planning
- Added unit tests and CI/CD pipelines using Bamboo for continuous integration

Guidance, Navigation & Controls Engineer Intern

May 2023 – August 2023

Space Dynamics Laboratory

- Implemented a UKF with range iteration and least-squares orbit determination methods using optical navigation
- Performed Monte Carlo analysis on relative orbits to identify challenging scenarios and refine estimation algorithms
- Developed unit tests for **Lambert Solver** integrated with **Initial Orbit Determination (IOD)** pipeline

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## RESEARCH PROJECTS

### Global Trajectory Optimization Competition (GTOC 6 & 11) – Modeling & Simulation

- Developed high-fidelity simulations for interplanetary trajectory design and optimization problems under nonlinear dynamics
- Contributed to solution verification and analysis workflows; maintained analysis notebooks and documentation for team reproducibility

### Delta-V Minimization from Geostationary Orbit to Mars

- Applied trajectory optimization techniques to minimize delta-v for an Earth-to-Mars transfer, improving fuel efficiency
  - Generated pork-chop plots using Lambert solutions and validated optimizer results against global minima
  - Visualized optimized trajectories and planetary motion via Blender's Python API
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## TEACHING & MENTORING EXPERIENCE

### Youth Development Professional

October 2024 – January 2025

*Boys & Girls Club*

- Designed and delivered an after-school **computer literacy** curriculum for grades 3–6 (typing, file systems, internet safety).
- Taught **introductory coding** via Scratch and beginner Python with project-based mini-units.
- Used quick pre/post checks and differentiated supports to boost engagement and skill growth.

### Math Teacher (Algebra I, Algebra II, Precalculus, Calculus)

October 2024 – January 2025

*North Star School*

- Developed **unit plans, assessments, and problem sets** aligned to course objectives; integrated Desmos/GeoGebra.
- Differentiated instruction and provided timely, standards-based feedback.
- Connected math to **real-world STEM** applications to deepen understanding.

### Research Assistant, AE 298 National Defense Education Program

January 2023 – May 2024

*Aerospace Engineering, University of Illinois Urbana-Champaign*

- Taught an introductory **rocketry** course for first- and second-year students, delivering lectures, labs, and build activities.
- Collected, cleaned, and analyzed course/IRB-aligned program data to evaluate student engagement, self-efficacy, and learning impact.
- Redesigned course structure (assessment pacing, project sequencing, and support materials) to improve retention and measurable learning outcomes.

### Research Assistant, Grants for Advancement of Teaching in Engineering (GATE)

November 2023 – May 2024

*Mechanical Engineering, University of Illinois Urbana-Champaign*

- Built a **final project framework** using interdisciplinary pedagogy to align hands-on rocketry with course objectives.
- Researched and authored a **literature review** on engineering-education/rocketry; contributed analysis and writing to a conference manuscript.

### Teaching Assistant, ASTR 122/210

August 2023 – May 2024

*Astronomy, University of Illinois Urbana-Champaign*

- Led engaging weekly discussion sections and developed comprehensive exam review materials for over **150** students.
- Created personalized study resources and efficiently graded **150+** assignments and exams, ensuring consistency and timely feedback.

### Graduate Career Peer

August 2022 – May 2024

*Engineering Career Services, University of Illinois Urbana-Champaign*

- Advised undergraduate and early graduate students on career planning, resumes/cover letters, and job/internship search strategies.
- Designed and led **mock interview** sessions (behavioral and technical) with targeted, actionable feedback to improve interview performance and confidence.

### Graduate Assistant, National Defense Education Program

January 2023 – May 2023

*Aerospace Engineering, University of Illinois Urbana-Champaign*

- Built partnerships with local **K–12** schools to deliver rocketry/avionics outreach, demos, and curriculum-aligned activities.
- Mentored teachers and students on safe build/testing practices and classroom integration; supported impact/data collection to document outcomes.

### Calculus, Physics, and Differential Equations Tutor

January 2019 – May 2022

*Academic Success Center, Iowa State University*

- Delivered customized lessons and problem-solving sessions for diverse learners in an inclusive, student-centered environment.
- Drove measurable academic gains, with average grade improvements of **5%–10%** across supported students.